



equipped with a robust library of physics models, enhance the precision of these simulations.

The next phase of battery modelling involves a suite of software tools and features offering simulations across a wide range of scales. This includes examining the microstructure of battery electrodes, where tools analyse 3D electrode morphology to predict the impact on electromechanical performance. These tools assist both material manufacturers and cell designers, enabling simulations from the microstructure level to cell modelling. They offer standard modules that users can customise for specific simulation needs.

Battery simulation also encompasses battery pack design, with tools analysing the complexity of the pack and its thermal management system. This holistic approach improves EV battery efficiency. Another crucial element is system integration, where co-simulation across different modules ensures cohesive results in designing efficient EVs.

A case study demonstrates the practical application of these tools. Faced with the challenge of implementing a simulation tool accessible to both specialised engineers and designers, a leading automotive company adopted a model-sharing approach. This reduced late-stage design changes and enabled precise motor sizing, contributing to an integrated mechatronics solution. The solution also aligned with programmable logic controller (PLC) programming, showcasing versatility across various aspects of battery manufacturing and design. This approach, embraced by industry leaders, highlights its success and influence on battery production and EV development.

The vital role of motors in vehicle electrification

A crucial element in the electrification of vehicles is the development and op-

Automotive industry embraces lighter materials and new technology

The integration of new materials and technologies is reshaping how vehicles are designed and built, marking a new era in automotive engineering.

Lightweighting in the automotive industry:

- Increasing use of materials like aluminium for vehicle structures.
- Focus on managing vehicle weight and enhancing efficiency.
- Use of robots for material handling and riveting, particularly on roof parts.

Rise of additive manufacturing:

- Gaining popularity in both industrial and commercial sectors.
- Essential for optimising component designs and sizes.
- Produces parts that match or exceed the strength of original designs.
- Particularly relevant in EV production.

Virtual commissioning in plant setup:

- Essential for planning new or upgraded manufacturing projects.
- Analyses machine performance, production lines, and integration of automated guided vehicles (AGVs) and human labour.
- Utilises simulation software for material flow, staffing, logistics, and energy management.
- Enhances manufacturing efficiency and effectiveness through pre-implementation scenario testing in a software environment.

As these technologies evolve, they are poised to become industry standards, redefining global automotive manufacturing. The shift towards innovative materials and processes is part of a wider move towards sustainability and efficiency in the industrial sector, aligning automotive production with environmental and economic objectives.

timisation of motors. Currently, there is a surge in research and development focusing on motor technology. One of the primary challenges in motor design is identifying areas of improvement to enhance motor performance. This includes designing efficient powertrains, a task that presents its own set of complexities.

Another significant challenge lies in managing the coupled thermal and electromagnetic conditions of motors. This aspect is critical whether dealing with low- or high-frequency motors. Additionally, addressing thermal issues in power electronics is paramount. Inadequate thermal performance optimisation can lead to severe consequences, as seen in instances where vehicles have caught fire due to poor thermal management.

To tackle these challenges, specialised tools such as electromagnetic simulation, thermal CMB stress analysis, vibroacoustic analysis, and

power electronics thermal simulation have been developed. A comprehensive suite of products exists that can perform these physics simulations, offering integrated solutions.

Among these tools, a simulation software focused on magnetic field simulation is instrumental in predicting the performance of motors, generators, sensors, transformers, actuators, and solenoids. This software is complemented by another solution that provides a complete design and analysis framework for permanent magnet and induction motors, facilitating efficient EV design.

By coupling these tools, automakers can resolve numerous challenges in motor design and optimisation. Leading automotive manufacturers are already using these advanced solutions for battery and motor management systems, showcasing their effectiveness and necessity in the fast-evolving realm of vehicle electrification. These tools